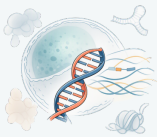
A

Evidence map: which information changes the recoverable signal?

INFORMATION LANDSCAPE



measured context → target evidence

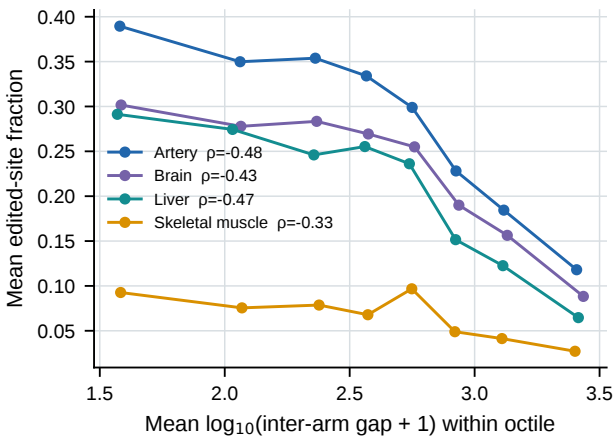


Tested formulation	Outcome	Interpretation
LEARNING & SUPERVISION		
Pair-first expanded supervision	● Selected	More labels; correct pair weight
Continuous + ordinal auxiliary labels	▲ Limited	Small ranking gain
Shared four-tissue training	▲ Limited	Transfer below preset criteria
TARGET REPRESENTATION		
Aligned target + helix-partner view	● Selected	Passes shifted-window control
Focal node / contact geometry	▲ Limited	Localizes target effect
Frozen RNA-language embeddings	✖ Not supported	F1-only; no ranking gain
Substrate-wide all-A decoder	✖ Not supported	Stopped for futility
Larger / deeper graph networks	✖ Not supported	No consistent benefit
Graph Transformer / GATv2 expansion	+ Not pursued	Simpler validated model sufficient
SUBSTRATE & CELLULAR CONTEXT		
Pair geometry / editing burden	◆ Supported*	Substrate propensity detectable
Hierarchical engagement gate	▲ Limited	Ranking gain; unstable F1
Bulk H3K36me3 overlap	▲ Limited	Liver AUPRC only
RBP binding × bulk expression	✖ Not supported	No residual gain
Transcript / repeat annotation panel	✖ Not supported	No site-level gain
Matched donor/cell multimodality	■ Deferred	Matched data unavailable
Matched in vivo structure / isoforms	■ Deferred	Matched data unavailable
INTEGRATION		
Fixed whole-duplex + local-augmented logits	● Selected	Complementary predictions
Unified end-to-end architecture	✖ Not supported	Below frozen ensemble

See Table S2 for comparators, effect sizes, and intervals.
*Reproducible observation not yet useful for site prediction; deferred hypotheses were not tested.

B

Geometry tracks substrate-level editing burden



C

Geometry improves pair-level burden models

